
Project Proposal - ECE 285

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Abstract

For this final project, we are trying to develop an object detection network. The purpose of this model is to allow a massage robot to identify different parts of the body. The network will be given an image or video and will need to be able to provide both the bounding box and its classification label. Our project will explore a dataset of posterior body images and implement two different approaches that allow real-time classification of different body parts given images of varying qualities or transformations.

1 Problem Definition

Hiring a masseuse is expensive and inconvenient, and a massage chair does not provide an actual massage experience. A robot can be used instead to perform these duties, but a method to identify relevant muscles and body parts is needed. Without a reliable method for the robot to "see", ordering a massage robot is no different from a massage chair.

As a result, a key necessity of the massage robot is its ability to quickly identify where certain muscles and body parts are in real-time. The importance of "real-time" becomes more evident when considering its requirement to be able to consistently produce accurate classification of different parts of the posterior with varied inputs. If the user moves, shifts or is slightly off-centered, the robot needs to still be able to identify the same regions. Likewise, the images the robot "sees" may become skewed, flipped or blurry at each frame, and the robot must be able to maintain accurate classification so the robot may remain reliable and safe.

2 Tentative Method

We plan to use the *You Only Look Once (YOLO)* and *Fast Region-based CNN (Fast R-CNN)* models to perform near real-time bounding box object detection. For both models, we plan to train on approximately 80%-90% of the pre-processed images to identify different body parts that should be massaged.

YOLO was chosen for its fast training and classification speed along with its more simple regression-like classification approach. This model trains on the entire image and learns more generalized features for each object, which is ideal for our use case in which noisy or unexpected inputs are still expected to generate accurate classifications. [2]

Fast R-CNN was selected for its similar relatively fast training and classification speed, but also for its regional classification approach. R-CNN by itself will be helpful for us in identifying specific regions of the posterior body. Fast R-CNN will improve the training time to allow for real-time identification of regional muscles and other body classifications. [4]

3 Experiments

We are planning on using the Posterior Body Image Dataset [3]. The dataset consists of 2,549 PNG images of the human back at different angles with 19 categories of different body parts. There are also approximately 25,000 total JSON files of bounding boxes available that can be used for validation.

The images will be train-test split with an initial ratio of 8:2 and will be adjusted accordingly. Additional random transformations and varying levels of blurring will also be applied to each image using OpenCV to simulate noisy or unexpected conditions attributed to movement or possible camera faults, which will simultaneously allow for more generalization.

Different candidate approaches using YOLO and Fast R-CNN must be explored along with tuning the hyperparameters. For YOLO, we plan to look into the performance of its different versions [5, 1]. We plan to start with around 10 convolutional layers and adjust accordingly. All other parameters and the choice of loss function will be set to the default provided by the model and adjusted accordingly. For Fast R-CNN, we plan to start with a traditional VGG CNN backbone and then explore the possible use of ResNet to increase the training speed. Like YOLO, the default parameters and loss functions will be explored first and then adjusted accordingly.

References

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- [4] Rohith Gandhi. R-cnn, fast r-cnn, faster r-cnn, yolo - object detection algorithms, Jul 2018.
- [5] Sik-Ho Tsang. Brief review: Yolov5 for object detection, Jun 2023.